

How the Room Occupancy System Works

A plain-language guide to how three sensors detect if someone is inside a room, how alerts escalate when they stay too long, and how we handle device disconnections.

The Three Sensors

The system uses three Zigbee sensors working together. No single sensor can tell the full story — it is the combination of all three that gives us a reliable picture.

Sensor	Type	What It Does
Door Sensor	Contact sensor	Detects when the door opens and closes. Tells us someone passed through the doorway.
Curtain Sensor	PIR motion (outside)	Mounted outside the door. Detects someone approaching or leaving the room entrance.
Motion Sensor (Master)	PIR motion (inside)	Mounted inside the room. Confirms someone is actually moving around inside.

Why three sensors? A door sensor alone cannot tell if someone walked in or just opened the door. The curtain sensor alone cannot tell if they entered the room or walked past it. All three together eliminate false positives.

How Entry Detection Works

The system does not mark the room as "occupied" from a single sensor. It waits until all three sensors confirm someone entered. The signals can arrive in any order — what matters is that all three happen within a 2-minute window.

1 **Curtain sensor fires** — someone walked up to the door from outside

Door opens and closes — someone passed through the doorway

3 **Master motion fires inside** — confirms a person is now inside the room

Once all three signals are received, the room is marked **OCCUPIED** and the family receives a "Room Occupied" notification. If only one or two sensors fire (say the door opens but nobody goes in), the system stays in "Entry Pending" and eventually resets — no false alarm.

How Exit Detection Works

When someone leaves, the system looks for another combination of signals:

- 1

Door opens and closes — someone passed through the doorway again
- Curtain sensor fires — someone walked away from the door outside
- 3

Master motion goes silent — no more movement detected inside the room

Once all three conditions are met, the room flips to **VACANT** and the family receives a "Room Vacant" notification.

The Four Room States

At any moment, the room is in one of four states:



State	Meaning
Vacant	Nobody is in the room. Waiting for entry signals.
Entry Pending	Some sensors fired but not all three yet. Waiting for the rest within the time window.
Occupied	Confirmed — someone is inside. Alert timers start running.
Exit Pending	Exit signals started. Waiting for confirmation that they actually left.

What Happens While Someone Is Inside

1. Stillness Alert

If the master motion sensor stops detecting movement for more than **1 minute** and no other sensor activity is seen, the system sends an emergency notification: "No movement detected — please check." This catches situations where someone might have collapsed or become unresponsive.

2. Long Stay Alerts (Escalating)

If someone stays in the room beyond normal durations, alerts escalate in three levels:

Level	Time	Severity	What Happens
Level 1	30 min	Warning	"Someone has been in the room for over 30 minutes."

Level 2	35 min	Critical	"Extended stay — please check in." Stronger push notification.
Level 3	40 min	Emergency	Full-screen P0 alert on the app. Requires acknowledgement from family.
Repeat	Every 5 min	Emergency	Keeps repeating with updated elapsed time until room becomes vacant or family acknowledges.

Night mode: The thresholds can be configured differently for nighttime hours (e.g., bathroom visits at 2 AM may have longer acceptable durations).

3. Safety Ceiling

If the room stays occupied for **4 hours** regardless of any other alerts, a safety ceiling alert fires. This is the absolute backstop — something is clearly wrong if someone has been in a room for that long.

Time Windows (Production Values)

Parameter	Value	Purpose
Entry window	2 min	All 3 entry signals must arrive within this window
Exit window	5 min	All 3 exit signals must arrive within this window
Exit silence timeout	1 min	How long master motion must stay silent to confirm exit
Stillness threshold	1 min	No-motion duration before stillness alert fires
Long stay L1 / L2 / L3	30 / 35 / 40 min	Escalating alert thresholds
Emergency repeat	Every 5 min	How often L3 emergency repeats after first fire
Safety ceiling	4 hours	Absolute maximum stay before forced alert

Test mode uses shorter timers (3 / 3.5 / 4 min for long stay, 30-second repeat) for quick walk-through testing.

How Data Flows — Sensor to Notification

1 **Sensor triggers** — Zigbee radio signal sent over the air at 2.4 GHz

ZBDongle-E (USB coordinator on the Raspberry Pi) receives the signal

3 **Zigbee2MQTT** software translates it into an MQTT message with sensor data

MQTT broker on the Pi distributes the message to subscribed services

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Backend service (room_occupancy_service) processes the state machine — decides if room is occupied/vacant

Cron job runs every 30 seconds (occupancy_alert_checker) — checks all occupied rooms against alert thresholds

7

Push notification sent to the family member's phone app + real-time socket event updates the dashboard

Device Disconnection and Auto-Reconnect

The Problem

All five Zigbee sensors in the system are battery-powered "sleepy" devices. They spend most of their time asleep to save battery and only wake up briefly to send data or check in with the coordinator (the USB dongle on the Pi). This means:

- If a sensor moves out of radio range (beyond ~10-15 metres indoors through 3-4 brick walls), it loses its connection to the coordinator.
- The coordinator has no way to "call" a sleeping sensor — it can only wait for the sensor to wake up and check in.
- Without monitoring enabled, there is no way to know a sensor has gone offline until you notice missing data.

What Causes Disconnection

Cause	Example
Out of range	Sensor placed too far from the coordinator, too many walls in between
Battery died	Battery voltage dropped below operating threshold
WiFi interference	WiFi router and Zigbee both use 2.4 GHz — channel overlap causes signal loss
Physical obstruction	Metal doors, RCC columns, or aluminium-framed partitions blocking the signal path
Coordinator restart	Zigbee2MQTT or the Pi restarted — sensors need to re-establish connection

The Solution — Availability Tracking

Zigbee battery sensors **automatically attempt to reconnect** when they come back into range. This is built into the Zigbee protocol — no manual re-pairing is needed. The sensor wakes up periodically (every 1-4 minutes depending on firmware), tries to reach its parent, and if successful, resumes normal operation.

To make this visible and actionable, we enabled **availability tracking** in Zigbee2MQTT by adding two settings to the configuration file:

Setting	Value	What It Does
availability	true	Z2M tracks each device's online/offline status. If a device misses its expected check-in, it is marked offline. When it checks in again, it is marked online.
advanced > last_seen	ISO_8601	Records a timestamp every time the coordinator hears from a device. Shows exactly when each sensor was last active.

Configuration location on the Pi: /home/pi/zigbee2mqtt/data/configuration.yaml — add "availability: true" at the top level and "last_seen: ISO_8601" under the existing "advanced:" section. Restart Zigbee2MQTT after saving.

What This Gives Us

- Real-time online/offline status for every sensor visible in the Z2M dashboard.
- "Last seen" timestamps showing exactly when each device last communicated.
- The backend can detect when a sensor goes offline and alert the family or maintenance team.
- When a sensor comes back into range, Z2M automatically marks it online again — no manual intervention.

Extending Range with a Repeater

If sensors are too far from the coordinator, a Zigbee repeater can be added to extend coverage. A mains-powered device like the **SONOFF ZBMINI-L2** smart switch acts as a Zigbee router — it relays messages between distant sensors and the coordinator, effectively doubling the range.

Metric	Without Repeater	With Repeater
Indoor range	10-15 metres	20-30 metres
Brick walls	3-4 walls	6-8 walls
Glass partitions	6-8 panels	12+ panels
Coverage	~2-3 rooms from Pi	Full 3 BHK flat

The repeater should be placed roughly halfway between the Pi and the farthest sensor, in an open spot (not inside a cupboard), on an inverter-backed socket, and at least 1 metre from the WiFi router.

Key Things to Know

- **Passive system** — the elderly parent does not need to do anything. No buttons to press, no app to use. The sensors work silently in the background.
- **Three-sensor confirmation** — all three sensors must agree before the room is marked occupied. This prevents false alarms from pets, curtain movement, or doors opening in the wind.
- **Privacy-first** — PIR motion sensors detect movement only. They cannot see through walls, identify who the person is, or record any video or audio.
- **Configurable per room** — alert levels can be set differently for each room and can have separate settings for daytime vs nighttime.
- **Auto-reconnect** — if a sensor temporarily loses connection (e.g., battery replaced, brief interference), it automatically reconnects when it wakes up. No manual re-pairing needed.
- **PIR limitation** — motion sensors cannot detect a person sitting perfectly still. That is what the stillness alert is for — if motion stops while the room is occupied, the system flags it.
- **Range limitation** — sensors must stay within radio range of the coordinator (or a repeater). Beyond 3-4 brick walls without a repeater, signal becomes unreliable.